

WHAT IS AN LLM?

A **Large Language Model (LLM)** is a type of artificial intelligence trained on vast quantities of text to predict and generate human-like language. Models such as Meta's **Llama 3** and Mistral AI's **Mistral 7B** contain billions of parameters encoding grammar, knowledge, and reasoning.

LLMs can be **steered via prompt engineering**: by constructing the input text carefully, developers control the model's persona, vocabulary, and knowledge boundaries—without retraining. This makes them ideal for powering dynamic NPC dialogue.

NPCs & DIALOGUE IN GAMES

A **Non-Player Character (NPC)** is any game character not controlled by the player. NPCs serve as merchants, guards, quest-givers, and civilians, making game worlds feel inhabited and alive.

Traditional dialogue uses **pre-authored branching trees**: the player picks from fixed options, each leading to a scripted response. This is reliable but fundamentally limited—every conversation must be hand-written, NPCs become repetitive, and players quickly exhaust all options.

THE PROBLEM

When LLMs are naïvely plugged into NPCs, **all characters sound identical** and know everything. A farmer discusses foreign diplomacy; a king uses casual slang. Individual identity and social hierarchy vanish entirely.

Research Question: How can LLM-driven NPCs be made to feel like distinct individuals with socially-bounded knowledge, unique speech patterns, and realistic awareness of world events?

The solution is a **knowledge-gated prompt pipeline**—middleware that constructs each NPC's context based on their social role, personal history, and proximity to events, so the LLM only receives what that NPC would realistically know.

SYSTEM ARCHITECTURE

The system uses a **decoupled middleware architecture**. Unreal Engine 5 sends player input and game state via HTTP/JSON to a Python FastAPI service. The middleware retrieves the NPC's profile, assembles a knowledge-gated prompt, queries a local LLM, and returns validated dialogue to UE5.



KNOWLEDGE GATING

The core innovation is a **five-tier knowledge hierarchy**:

- Tier 0 – Personal:** Own experiences, family, daily life.
- Tier 1 – Local:** Village events, nearby geography, gossip.
- Tier 2 – Regional:** Town politics, trade routes, conflicts.
- Tier 3 – National:** Wars, royal decrees, alliances.
- Tier 4 – Scholarly:** History, magic theory, foreign lands.

A peasant (Tier 0–1) knows village life and gossip. A noble (Tier 0–4) knows politics and history. The middleware **only injects tier-appropriate information** into the prompt—the LLM physically cannot reference restricted knowledge.

NPC PROFILES & SPEECH

Each NPC has a **JSON profile**: name, archetype, knowledge tiers, personality, speech directives, backstory, and relationships.

Speech fingerprinting enforces distinct voices through vocabulary and grammar constraints. A peasant uses dialect and simple sentences; a noble uses formal register and courtly phrasing. Two farmers still feel like different people, while a farmer and noble are immediately distinguishable by **how** they speak.

EVENT PROPAGATION

World events propagate through an **event graph** with realistic delays. **Distance:** nearby NPCs learn quickly; distant ones hear rumours or nothing. **Social role:** nobles get messengers; merchants carry trade-route news; peasants rely on hearsay. **Time decay:** details degrade as information travels.

HYPOTHESES

H1 – Believability: Players will rate LLM-driven NPCs as significantly more believable and immersive than dialogue-tree NPCs, measured via Likert-scale questionnaire.

H2 – Knowledge Boundaries: The gating system will prevent out-of-tier information leakage in ≥95% of adversarial queries.

H3 – Distinctiveness: Blind-test participants will identify NPC social class from transcripts alone at >60% accuracy.

H4 – Latency: Local inference will achieve median response time under 2 seconds with a quantised 7B model.

TESTING & DATA COLLECTION

User Study (H1, H3): Participants interact with both LLM and static-tree NPCs in counterbalanced order. Post-interaction Likert questionnaires measure believability, individuality, and immersion. A blind test presents transcripts for archetype identification.

Adversarial Probing (H2): 100+ queries designed to trick NPCs into revealing restricted information. Each response scored pass/fail.

Performance Benchmarks (H4): 500+ automated exchanges measuring end-to-end latency, VRAM usage, and token throughput. Quantised models (Q4, Q5) compared against full-precision.

DIALOGUE COMPARISON

PEASANT (Tier 0–1)

Player: "What do you know of the war?"
"War? I heard soldiers marchin' through last moon. Took old Brin's cart and never paid. That's all I know — we keep our heads down."

NOBLE (Tier 0–4)

Player: "What do you know of the war?"
"The eastern alliance fractured after Duke Varren's betrayal at Thornwall. Our forces hold the river, but supply lines thin. Speak of this to none."

OUTCOMES & FUTURE WORK

Knowledge-gated prompting is predicted to produce NPC dialogue perceived as significantly more believable than static trees, while maintaining reliable information boundaries and viable latency. The key contribution is a **reusable architectural pattern** for integrating LLMs into game NPCs with socially-realistic constraints.

Future Work: Memory persistence across sessions; emotional state modelling affecting tone; inter-NPC gossip creating emergent narrative; voice synthesis with archetype-distinct voices; fine-tuning on fantasy RPG dialogue for improved consistency.